Ngoc Nguyen

(646) 456-3821 | nguyengo@dickinson.edu | LinkedIn | Portfolio

Education

Dickinson College Carlisle, PA
B.S. Data Analytics, Finance & Quantitative Economics (Triple Major) | Dean's List *Expected May 2028*

- **Relevant Coursework:** Data Systems, Data Visualization, Machine Learning, Probability & Statistics, Econometrics
- **Certifications:** DataCamp (Intermediate SQL; Power BI & DAX); Goldman Sachs Job Simulation (Forage)

Professional Experience

Transfer Pricing Intern May 2026 – Aug 2026
BDO Vietnam *Hanoi, Vietnam*

- Translated 2,000+ company descriptions from English to Vietnamese with **Python** and the **OpenAI API**, with caching and checkpoints so interrupted runs could resume, **cutting review time from 5+ days to about 1 hour**.
- Screened a **BvD Orbis** data set of 10M+ companies against 9 selection criteria, narrowing down to final sets of about 10 comparable companies per search across 3 client engagements in logistics, steel manufacturing, and electrical equipment.
- Built exclusion matrices and interquartile ranges of comparable companies' operating margins, and checked whether each client's margin fell within that range.

Data Analytics Intern May 2025 – Aug 2025
VNPay – Vietnam Payment Solution JSC *Hanoi, Vietnam*

- Queried and analyzed **25.2M+ movie-ticket transactions** from 1M+ customers in **SQL** and **Python** to pinpoint peak sales windows and the top-performing genres within each peak.
- Developed an **RFM segmentation model** with 5 value tiers, finding 41% of customers loyal or high-value and 38% lapsed; the tiers guided the team's retention campaigns, and **that share rose to over 50% in 3 months**.
- Segmented cinema customers into 4 groups using **K-means clustering** on genre preferences and purchase timing, then recommended targeted offers such as weekend family bundles and ride vouchers.
- Presented insights and recommendations to team leads and the marketing and sales teams.

Project Experience

Flight Delay Prediction, Harrisburg International Airport (MDT) Portfolio
Python (pandas, scikit-learn), BTS On-Time Performance data

- Cleaned and explored 29K+ BTS flight records (2020–2025) departing MDT, and engineered features such as departure hour and previous-flight delay to predict arrivals 15+ minutes late.
- Found that delays build up through the day, from about 7% of 5 a.m. departures to over 40% after 9 p.m., and that flights whose previous leg was delayed arrived late **70% of the time vs. 17%** otherwise.
- Compared **Logistic Regression** and **Random Forest** classifiers; the Random Forest scored the higher ROC-AUC (0.67 vs. 0.65) and caught 56% of delayed flights.

Maritime Trade Risk Dashboard Portfolio
Python, SQL Server, Power BI, DAX, UN Comtrade API

- Engineered an end-to-end **ETL pipeline** that pulls 284K+ rows from the UN Comtrade API with **Python** and loads them into a **SQL Server star schema**.
- Built a 7-page **Power BI** report with **DAX** risk-score measures that drills down from category to product.
- Identified 272 bulk products accounting for 64% of ocean freight value but unsuitable for **air freight**, and uncovered a data error that reversed the original report's conclusion.

U.S. Gender Wage Gap, 2012–2025 Portfolio
Python (pandas, statsmodels, pyfixest), Stata

- Conducted 42 survey-weighted fixed-effects regressions in **Python** on 644K workers, showing the adjusted gap has held near 21% since 2012, and validated every result by reproducing it in **Stata**.

Leadership and Involvement

Lead Community Service Aide Aug 2026 – Present
Dickinson College Department of Public Safety *Carlisle, PA*

- Recruit and train student aides, and lead building security checks and audits of campus lighting and emergency phones.

Vice President, Vietnamese Student Association Sep 2025 – Sep 2026
Dickinson College *Carlisle, PA*

- Led cultural festivals and events for 200+ students and partnered with college departments on sustainability projects.

Skills

- **Programming:** SQL (SQL Server, SQLite), Python (pandas, scikit-learn, matplotlib, seaborn, statsmodels), R, Stata
- **Data & BI:** Power BI (DAX, Power Query), Tableau, Excel, Qlik Cloud, Git
- **Analytics:** ETL, star-schema modeling, customer segmentation (RFM, K-means), regression (OLS, logistic, quantile)
- **Languages:** Vietnamese (native), English (fluent)